Fiscal Year:	FY 2019	**Task Last Updated:**	FY 06/07/2019
PI Name:	Blanco, Gustavo Ph.D.		
Project Title:	Space Flight-Altered Motility Activation and Fertility-Dependent Responses in Sperm from Sea Urchin and Rodents		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate (3) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	66160-0001	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	02/09/2018	**End Date:**	03/31/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 3/31/2020 per F. Hernandez/ARC and NSSC information (Ed. 5/15/19) NOTE: Extended to 8/31/2019 per F. Hernandez/ARC (Ed. 10/29/18)		
Key Personnel Changes/Previous PI:	January 2018: Former Principal Investigator was Joseph Tash, Ph.D., who retired January 2018. Dr. Tash remains as consultant on the project.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC18K0522		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE (June 2018): Continuation of "Space Flight-Altered Motility Activation and Fertility-Dependent Responses in Sperm from Sea Urchin and Rodents" (grant number NNX13AM62G) due to retirement of former Principal Investigator Joseph Tash, Ph.D. Originally the grant had continued with same grant number as noted in January 2018 below note; in February 2018 new grant number 80NSSC18K0522 with new period of performance was awarded. NOTE (January 2018): Continuation of "Space Flight-Altered Motility Activation and Fertility-Dependent Responses in Sperm from Sea Urchin and Rodents" with the same grant number NNX13AM62G due to retirement of former Principal Investigator (PI) Joseph Tash, Ph.D. NOTE also that in 2016 with predecessor PI Tash, work on the grant was amended to include cryopreserved bovine and human semen, while eliminating the use of sea urchin and rat species. Sperm motility and acquisition of the capacity to fertilize are the most fundamental components of male reproduction necessary for species survival on Earth or in space. Previous space shuttle flight experiments that we performed demonstrated that the signal transduction (i.e., m protein phosphorylation) required for activation of sea urchin sperm occur more rapidly in microgravity than at 1G. In contrast, we found that signaling in sperm triggered by egg chemotactic factors is delayed in microgravity compared to 1G. No flight-capable microscope was available during those experiments to determine whether the signaling changes were in fact coupled to differences in motility under the microgravity versus 1G conditions. Our hypothesis is 1) that sperm motility (sea urchin and mammalian) is activated more rapidly in microgravity than on Earth, and 2) that the response of sea urchin and mammalian sperm (motility hyperactivation, and morphological changes) to fertilization activating conditions, as well as the underlying metabolic and regulatory mechanisms, are delayed or blocked in microgravity. In this regard, if the corresponding activation and changes in sperm motility do not occur, then fertilization cannot occur and reproduction will fail. Thus, it is essential to assess whether the critical component of sperm function, motility, and the underlying signalling and metabolic support are altered in space flight conditions. We propose to use existing flight hardware to activate sea urchin and rodent sperm in microgravity, and quantify multiple parameters of motility and morphology, combined with detailed metabolomic pathway analysis. Previously utilized methods for preparing inactive sea urchin sperm for subsequent activation in space flight will be adapted for mouse and rat sperm. We will also test whether activating peptides from sea urchin eggs, or mammalian capacitating conditions trigger temporal or magnitude changes in sperm motility, morphology, as well as essential regulatory glycolytic and energy metabolic pathway markers (using novel metabolomic analysis) in microgravity versus 1G. Most importantly, this comprehensive project will lead to the first direct in vitro assessment of canonical fertility-critical functional processes of mammalian sperm in microgravity that are identical to many human in vitro clinical and research laboratory assessments. These experiments are 1) part of the Rapid Turn-around Space Flight Experiments Research Emphasis, and 2) address the National Research Council (NRC) 2011 Decadal Survey to discover and characterize underlying mechanisms of adaptation to the space flight environment, and to determine cellular and organismal mechanisms that regulate and sustain reproduction and development in space. In this regard, the results of these experiments will provide important fundamental reproductive biological data to support NASA's goal to conduct multigenerational studies of mammals in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Previous studies indicate that reproductive functions in sperm that occur on the ground also occur in space flight. However, the timing of signal transduction changes upon initiation of sperm motility is more rapid, yet, the fertility-related responses of sperm to the presence of egg chemotactic factors are delayed. This suggests that sperm fertility may be affected resulting in subfertility or infertility in males during and post flight. There is insufficient data available from humans to assess this possibility due to low n's and HIPAA concerns.
	ED. NOTE (5/16/19): Compiled from PI's annual technical report submitted to NASA Ames Research Center in March 2019; covers period from December 2017-March 2019. As noted previously, the original specific aims were changed from rodent (mice or rat) and sea urchin to human and bovine sperm, with approval of the Space Biology Program. The species change was due to our findings that rodent and sea urchin sperm could not meet the logistics requirements for the time windows of SpaceX pre-launch to experiment activation, whereas human and bovine sperm were shown to meet these requirements. Hence, there is disparity between the species named in the grant title and the species actually used, targeted in the hypothesis and specific aims. The overall hypothesis driving the project is 1) that sperm motility (human and bovine) is activated more rapidly in microgravity than on Earth, and 2) that the response of mammalian sperm (motility hyperactivation, and morphological changes) to fertilization activating conditions, as well as the underlying signal transduction regulatory pathways (protein phosphorylation), are delayed or blocked in microgravity. Specific Aim 1: Determine whether the activation of human and bovine sperm motility in µG is more rapid and produces higher velocity parameters compared to 1G. The hypothesis for this aim is that sperm motility activates more rapidly in µG, and produces higher sperm velocity values than at 1G. This aim will fill the knowledge gap from our STS-81 and STS-84 flight experiments whether altered sperm motility is coupled to protein phosphorylation changes that we found earlier in µG. Specific Aim 2: Determine if capacitation of human and bovine sperm is inhibited in µG. The hypothesis for this aim is that the changes in sperm motility, protein phosphorylation, and acrosome status that are elicited during incubation under capacitating conditions will be delayed or blocked in µG compared to 1G. Achievements towards these goals: 1. After approval for flight, preparations for flight and asynchronous ground controls were completed. Flight kits and samples, and matched ground control kits and samples were completed at Kennedy Space Center (KSC) for launch on Space-X 14 on April 2, 2018. 2. Human samples were prioritized as more valuable than the bovine (scientific impact and significance). 3. Therefore, bovine samples were done first to maximize crew muscle memory and crew coordination of tasks prior to starting human samples. 4. A seventh bovine sample (B0) was provided for the crew to do first.

Task Progress:	5. Cryopreserved human and bovine sperm stored at the conditions of flight and in matched temperature condition ground control freezers at KSC displayed excellent recovery of motility. 6. The very first sample processed in flight (B0 bovine sample) served as a “training” sample for the crew. This sample proved to be absolutely vital for the subsequent success of the crew operations. This sample was the first time that both crew members worked together on the samples, even though they each had prior training separately on their own steps of the procedure. Major activities Flight kits and samples, and matched ground control kits and samples were completed at KSC for launch on Space-X 14 on April 2, 2018. About a week before launch of Micro-11, the onboard VUE camera system failed. The VUE system is required to operate the camera attached to the flight Microscope system. All attempts by Marshall Space Flight Center to repair the system failed. Since pre-flight operations established that the frozen sperm and activation media were stable for at least 9 months, the launch of samples was not impacted, because they could be stowed on the International Space Station (ISS) under proper conditions of temperature until the camera system was replaced. Therefore, the start of crew operations for Micro-11 was delayed until the broken VUE system was replaced and validated for resumption of experiment activities. The start of crew operations for Micro-11 were delayed until late June, and continued through July - Aug 2018. All bovine and human samples were processed. However, the new VUE R1 system also failed during Micro-11 bovine OPS causing loss of microscopy for the T20 timepoint in bovine sample B2, and all of the bovine sample B4 except for the T60 timepoint. This resulted in an n=2 for non-capacitation bovine samples instead of n=3 for all other samples. Immediately after the VU-R1 failure, Micro-11 operations were halted until a resolution to the VUE R1 failure could be resolved and assurances made to minimize additional failures and further loss of motility data for the remainder of the Micro-11 procedures. Revisions were made and no additional failures occurred. It should be noted that the loss of the bovine motility videos did not impact the MSG processing of any of the bovine sample fixation steps. Thus, we have a complete set of fixed bovine sperm samples (flight and ground), as well as a complete set of fixed humans samples (flight and ground) to be analyzed. In addition, there is the complete set of EVT (experiment verification test) ground samples (bovine only) that were performed using the same sperm sample aliquots that were used for flight and asynchronous ground controls at KSC. Samples were returned to Earth on Space-X 15 on Aug 3, 2018. Return of the flight samples allowed us to run the complete set of the asynchronous ground controls at KSC Aug 12 - 18, 2018. Asynchronous ground controls utilized the same pooled human and bovine sperm, activation media, and fixatives that were processed for flight and stowed at KSC under the same conditions. Initial sample analysis has focused on the human samples. The first data subset completed is the T30 timepoint of human sperm activation. This timepoint was selected because prior ground controls studies demonstrated that 30 minutes post-activation represented the earliest achievement of peak activation of motility and largest differences between capacitating (fertilizing) and non-capacitating (non-fertilizing) conditions. Conversion of all the remaining human sperm motility videos (flight and ground, capacitating and non-capacitating, at all timepoints) to SpermVision format has now been completed and the analysis of the motility is now being compiled for statistical analyses. 1. Initial analysis of Micro-11 human sperm motility data demonstrates that at T30 there are significant differences in several motility parameters known to be critical for sperm fertility. This finding provides strong data in support Specific Aim 1. However, this initial data suggests that sperm motility may be inhibited rather than increased in microgravity. Conversely, the initial analysis focused on capacitating conditions, also support Specific Aim 2, and the underlying hypothesis that changes in sperm motility elicited during incubation under capacitating conditions will be delayed or blocked in µG compared to 1G. 2. The ability to activate clinically cryopreserved human sperm in space flight has been achieved. 3. The quality of the sperm samples (both human and bovine) were outstanding as revealed by the video microscopy. 4. The sperm video microscopy collected in flight was proven to be quantifiable and validated using the SpermVision system. 5. With the completion of the human video sample conversions and SpermVision analysis going, we will analyze the bovine sperm motility samples that will best provide quantitative assessments of the impacts of the loss of the bovine video microscopy files and additional thaw-freeze cycle for flight sample B4.
Bibliography Type:	Description: (Last Updated: 04/23/2021)
Abstracts for Journals and Proceedings	Tash JS, Yarns EM, Piper SS, Ngo L, Laundry E, Julo M, Stodieck LS, Zea L, Karouia F, Blanco G. "Micro-11: Human and bovine sperm function in microgravity on the ISS." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Nov-2018
Abstracts for Journals and Proceedings	Tash JS. "Crew operations video for Micro-II MSG OPS: First human sperm sample." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Nov-2018